

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Text Structure and Purpose	Easy

Question

The majority of plastics today wind up in landfills or are, at best, recycled into materials that have a very limited range of applications. To address this problem, chemist Guoliang Liu and colleagues designed a reactor that melts polyethylene and polypropylene—two widely used plastics—into a wax. The wax can then be transformed into a surfactant (a chemical compound usable as a detergent). With this promising new method, plastic waste could be turned into a range of useful cleaning products.

Which choice best states the function of the underlined portion of the text?

Answer

- A. It clarifies the meaning of a scientific term.
- B. It describes an environmental concern.
- C. It explains the significance of a scientific discovery.
- D. It identifies a result that confused the team.

Correct Answer: A

Rationale

Choice A is the best answer because it most accurately describes how the underlined phrase functions in the text as a whole. The text states that the wax produced by Liu and colleagues’ reactor can be turned into a surfactant. The underlined phrase, which is set off with parentheses, then provides a definition for the term "surfactant," explaining that it’s a chemical compound that can be used as a detergent. Thus, the underlined portion of the text functions to clarify the meaning of a scientific term.

Choice B is incorrect. Though the text as a whole focuses broadly on an environmental concern, the underlined phrase does not; it simply indicates what a surfactant is. Choice C is incorrect. Though the text as a whole focuses on a scientific discovery (Liu and colleagues’ solution to the problem of plastic recycling), the underlined phrase does not explain its significance; it simply defines a scientific term used in the discussion. Choice D is incorrect. Though the text as a whole includes discussion of the result found by Liu and colleagues, the underlined phrase does not discuss it; it simply defines a scientific term used in the discussion. Additionally, at no point in the text is it mentioned that the team was confused.